Phase II: *Exploratory analysis and hypotheses.* Due Oct. 15 for peer review, and Oct. 20 for final grading

Part 1: *Data collection and exploratory data analysis.*

- Settle on a single idea and state your research question(s) clearly.
- Carry out most of your data collection and cleaning.
- Compute some relevant summary statistics, and show some plots of your data, as applicable to your research question(s). Use this exploratory data analysis to:
    - update your research question(s), if applicable;
    - update your data description, if applicable (e.g. if you collect additional data).
- **To receive credit for Phase 2 Part 1, you must submit a PDF of an executed Jupyter notebook** on Gradescope with the following sections:
    - **Research question(s).** State your research question (s) clearly.
    - **Data collection and cleaning.** Have an initial draft of your data cleaning appendix. Document every step that takes your raw data file(s) and turns it into the analysis-ready data set that you would submit with your final project. Include text cells describing your data collection (downloading, scraping, surveys, etc), and text cells describing any additional data curation/cleaning (merging data frames, filtering, transformations of variables, etc). Include code cells for data curation/cleaning, but not collection.
        - Note: You should be saving data in intermediate files at several points through this process so that you are not starting from scratch every time you change something. This also makes sure you have documentation for everything that you have done for your reference, but we do not necessarily need to see all of it in code form.
    - **Data description.** Have an initial draft of your data description section (details below). Your data description should be about your analysis-ready data.
    - **Data limitations.** Identify any potential problems with your dataset.
    - **Exploratory data analysis.** Perform an (initial) exploratory data analysis.
    - **Questions for reviewers.** List specific questions for your peer reviewers and project mentor to answer in giving you feedback on this phase.

Data description

- This should be inspired by the format presented in Gebru et al., 2018. Answer any relevant questions from sections 3.1-3.5 of the Gebru et al. article, especially the following questions:
    - What are the observations (rows) and the attributes (columns)?
    - Why was this dataset created?
    - How did you create this dataset? What tools, procedures, and data sources did you use?
    - What processes might have influenced what data was observed and recorded and what was not?
    - What preprocessing was done, and how did the data come to be in the form that you are using?
    - If people are involved, were they aware of the data collection and if so, what purpose did they expect the data to be used for?
    - Where can your raw source data be found, if applicable? Provide a link to the raw data (hosted on Github, in a Cornell Google Drive or Cornell Box).

Part 2: *Preregistration of analyses.*

- **Preregister** at least two analyses that you vow to present in your final report, no matter the results (significant or not).
    - Each preregistration statement should include 1) a hypothesis and 2) an analysis that will be run to support that hypothesis.
    - *Do not perform these analyses, check they are significant, and then "preregister" them.* That defeats the purpose. You can learn a lot from associations that are not statistically significant!
    - Keep in mind that for your final report, you must provide at least one model showing patterns or relationships between variables that addresses your research question.
- Include a "**Questions for reviewers"** section at the end of your submission, listing specific questions for your project mentor to answer in giving you feedback on this phase.

Submission checklist: To receive credit for *Phase 2 Part 2*, you must submit a .pdf through Gradescope with the following sections:

- **Preregistration Statement 1.** Must contain 1) a hypothesis and 2) a proposed analysis.
- **Preregistration Statement 2.** Must contain 1) a hypothesis and 2) a proposed analysis.
- **Questions for Reviewers.** List specific questions for your project mentor to answer.

Preregistration Example

- **Hypothesis 1:** Chocolate-based flavors are more popular on the East Coast than in the Midwest.

Analysis: Run a linear regression where we input region (as a dummy variable) and output proportion of chocolate-based flavors (# chocolate-based scoops / # total scoops) sold on an average day. The Midwest dummy will be our reference variable, so we will test whether $\beta_{eastcoast} > 0$.

- **Hypothesis 2:** The cost of ice cream increases with distance from Ithaca, NY.

Analysis: For data where each row represents a different location, we run a linear regression where we input the distance of the location from Ithaca, NY, and output price (in dollars) of ice cream. Because our distance measurement will have signed values (e.g. anything West of Ithaca would be negative miles and East would be positive miles), we will test whether $\beta_{distance} \neq 0$.

- **Questions for reviewers:**
    - Is our data set sufficiently balanced between urban/rural, and by region?
    - Are there other input features that could be relevant for the price of ice cream?
    - Does having a signed distance measurement metric make sense?

Note: To receive credit for *Phase 2*, you must submit **a PDF of an executed Jupyter notebook** through Gradescope. The two files from Part 1 and Part 2 should be combined and submitted together on Gradescope.

Rubric				
Category	**Least developed projects** *0 points*	**Less developed projects** *1 point*	**Typical projects** *2 points*	**More developed projects** *3 points*
Research question(s)	Research question(s) are missing.	Question(s) are not clearly stated or significantly limit potential analysis.	Clearly states the research question(s), which have moderate potential for interesting analyses.	Clearly states complex research question(s) that leads to significant potential for interesting analyses.
Data cleaning	No data cleaning has been performed.	Data is minimally cleaned, with little documentation and description of the steps undertaken. The data is not necessarily complex, with neither web-scraping performed nor use of multiple meaningfully different data sources. The dataset is nearly unchanged from when	Completes all necessary data cleaning for subsequent analyses. Describes cleaning steps with some detail. The dataset is likely complex, involving merging multiple data sources or performing web-scraping (but if not, it is clear that more work has been done in the EDA section).	Completes all necessary data cleaning for subsequent analyses. Describes all cleaning steps in full detail, so that the reader has an

		it was obtained from its original source.		excellent grasp of how the raw data was transformed into the analysis-ready dataset, to the extent that the reader can easily replicate the cleaning steps on their own with the raw data.
Data description	Data description is missing.	Simple description of some aspects of the dataset, little consideration for sources. The description is missing answers to applicable questions detailed in the "Datasheets for Datasets" paper.	Answers all relevant questions in the "Datasheets for Datasets" paper.	All expectations of typical projects + answers are all thorough, and credit and value the data sources.
Data limitations	Data limitations are missing.	The limitations are not explained in depth. There is no mention of how these limitations may affect the meaning of results.	Identifies potential harms and data gaps, and describes how these could affect the meaning of results.	Creatively identifies potential harms and data gaps, and describes how these could affect the meaning of results, and the impact of results on people. It is evident that significant thought

				has been put into the limitations of the collected data.
Exploratory data analysis (EDA)	No analysis is conducted.	Motivation for choice of analysis methods is unclear. Does not justify decisions to either confirm / update research questions and data description. Performs "chart barf": instead of thinking critically about which analyses to perform, simply produces iterations of all possible charts with little to no explanation.	Sufficient plots and summary statistics to identify typical values in relevant single variables and connections between pairs of relevant variables. Uses exploratory analysis to confirm/update clear research questions and data description.	All expectations of typical projects + analysis methods are carefully chosen (no "chart barf"!) to identify important characteristics of data.
Readability *(Note: if your group does not include a native English speaker, make a note of that in your submission)*.	The response is confusing or hard to read.	The response uses complete sentences, but is formatted without headings or without key points highlighted. Code cells have sparse comments or are missing comments entirely.	The response uses complete sentences and states every point clearly and simply. Code cells generally have comments and markdown cells explaining their purpose, but some may be missing. Most sections have a heading, and bolding and italics are used to highlight key points.	The response uses complete sentences, states every point as clearly and simply as possible (using bullet points or lists if helpful), and each cell of code has a comment and a markdown cell explaining its purpose. Each section has a heading, and bolding and italics are used to highlight key points.

Preregistration statement (x2)	The preregistration statement is missing. No hypothesis or proposed analysis is provided.	The preregistration statement is incomplete (i.e., missing a hypothesis or missing a proposed analysis). Both the hypothesis and proposed analysis are provided, but the ideas are generic and not closely related to the topic of the project.	Both the hypothesis and proposed analysis are provided. The ideas relate to the group's topic and research questions. The hypothesis is understandable and somewhat answerable with the data analysis proposed.	Both the hypothesis and proposed analysis are provided. The ideas demonstrate thoughtful engagement with the group's topic and research question. Concepts are clearly defined and, when relevant, claims are supported by evidence from prior research or domain knowledge. The hypothesis is well-informed and clearly answerable with evidence from the proposed analysis. The proposed analysis addresses the hypothesis and can be feasibly conducted using concepts and tools from course material.

Questions for reviewers	There are no questions for reviewers included.	The questions for reviewers are vague or unclear.	The questions for reviewers are specific to the datasets and are based on group discussions between team members.	The questions for reviewers are specific to the datasets and are based on group discussions between team members. Questions for reviewers look toward the next stage of the project.